EEE4114F Final Exam

June 2023

Duration: 3 hours | Total: 80 marks

Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has *4* questions totalling 40 marks. You must answer all of them.
- **Part B** has *4* questions totalling 40 marks. You must answer all of them.
- Part A and Part B must be answered in separate exam scripts.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 3 hours.

Part A:

Question:	1	2	3	4	Total
Points:	10	10	10	10	40
Score:					

Part B:

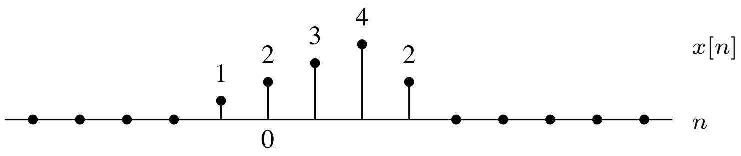
Question:	5	6	7	8	Total
Points:	8	7	12	13	40
Score:					

Part A: Digital Signal Processing

Parts A and B must be answered in separate exam booklets.

Question 1 [10 marks]

Suppose $x[n]$ is the signal below:



Plot the following:

a. $y_1[n] = x[2n-1]$

b. $y_2[n] = x[-n+1]$

c. $y_3[n] = x[n] - x[n-1]$

d. $y_4[n] = \sum_{k=-\infty}^{n} x[k]$.

Question 2 [10 marks]

Find the time-domain signals corresponding to the following transforms:

a. $Y(e^{j\omega}) = 12je^{-j6\omega}\sin(4\omega)$

b. $X(z) = z^2 + 3z + 5 + \frac{2}{z^2+5z+4}$ with ROC $|z| > 4$.

Question 3 [10 marks]

a. Consider $x_a(t) = \cos(100\pi t)$. Determine the minimum sampling rate (in Hertz) to avoid aliasing.

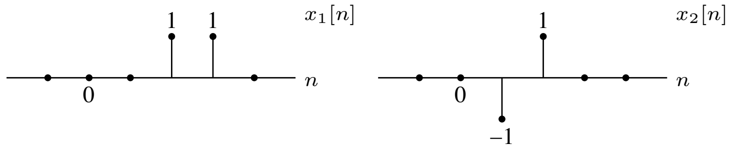
b. Write down an expression for $x_b[n]$ if a sampling frequency of 200 Hz is used on the signal $x_b(t) = \cos(100\pi t)$.

c. Plot the DTFT, over the range -2π to 2π, of the signal $x_c(t) = \cos(100\pi t)$ sampled at a frequency of 75 Hz.

d. If the sampling frequency is 48 kHz, what is the frequency of the discrete time signal corresponding to a sinusoid at 1.2 kHz?

Question 4 [10 marks]

Consider the signals below, and assume values of zero outside of the domain shown:



a. Find and plot the linear convolution $y_1[n] = x_1[n] * x_2[n]$.

b. Suppose $y_2[n] = x_1[n] \textcircled{N} x_2[n]$ is the 4-point circular convolution. If $Y_2[k]$ is the 4-point DFT of $y_2[n]$ then show that

$$Y_2[0] = 0, \quad Y_2[1] = -2j, \quad Y_2[2] = 0, \quad \text{and} \quad Y_2[3] = 2j.$$

c. Find and plot $y_2[n]$ over the range $n = 0$ to $n = 10$.

END OF PART A

Part B: Machine Learning

Parts A and B must be answered in separate exam booklets.

Question 5: Model evaluation [8 marks]

Figure 1 shows the output of three models used to predict the yield of a crop based on average soil moisture. All of the models were trained on the same data.

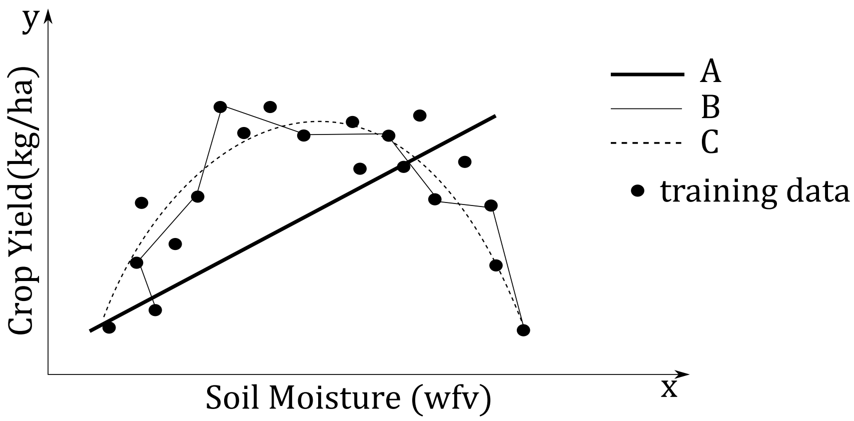


Figure 1: The output of three models trained to predict crop yield from soil moisture

a. Would it be possible to improve the performance of Model A by training with additional data? Explain your answer. (2)

b. If Model B was trained on additional data would you expect the output to be the same or different than what is shown in Figure 1? Is this a sign of high or low variance? (2)

c. What technique could have been used to produce Model C if it consists of the same structure and number of parameters as Model B? Explain your answer. (2)

d. Suppose the models are trained multiple times using random splits of the dataset into training and test sets. What issues could arise when using these results to evaluate models? (2)

Question 6: k-Nearest Neighbours [7 marks]

You are given the task of classifying fruits into two classes, lions and tigers, based on two features x_1 and x_2. You decide to use a k-Nearest Neighbour (kNN) algorithm to classify the fruits using Euclidean distance as the distance metric. Figure 2 shows the training samples as well as two test samples. You can consider lions to be class '0' and tigers to be class '1' to make it simpler to write. Answer the following questions.

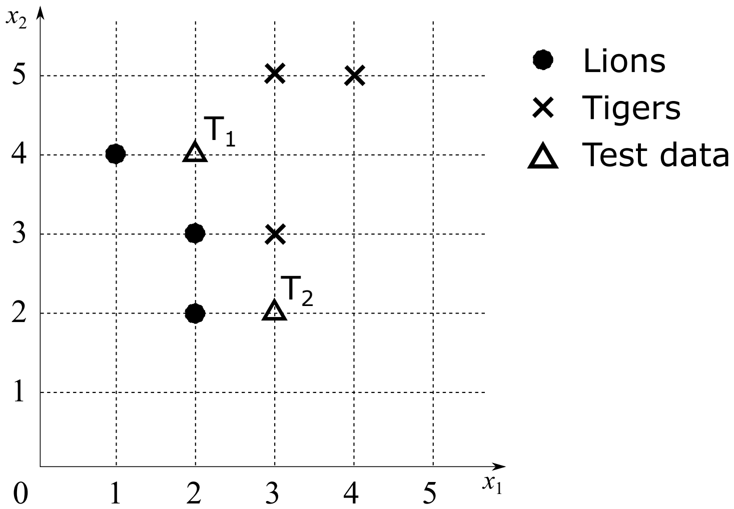


Figure 2: A plot of training and test examples with respect to features x_1 and x_2.

a. Does this nearest neighbour model have trainable model parameters? If so, what are they? (2)

b. If $k = 3$, how would the kNN algorithm classify each test point, T_1 and T_2. Show all working out and steps that the kNN would take. (4)

c. What problem could be caused by a value of k that is too small? (1)

Question 7: Neural Networks [12 marks]

Suppose you have a fully-connected, feedforward neural network with the weights initialised as shown in Figure 3. Each neuron's activation functions are Rectified Linear Units (ReLU). The input data has one feature and the target variable has two features.

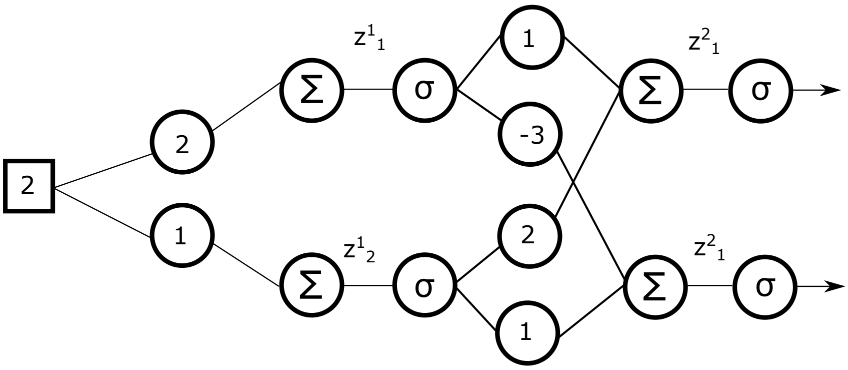


Figure 3: A fully-connected, feedforward neural network

The cost function you plan to optimize is given by: $J = \frac{1}{2}(\hat{y} - y)^2$

Where $\hat{y}$ is the predicted output of the model and y is the target variable.

The model is initialised with the bias values that have been omitted from the diagram for simplicity purposes only.

$$\mathbf{b}^1 = \mathbf{b}^2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

a. Suppose the input is $x = 2$, and the target value is given by $y = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

(i) Calculate the following values based on a single forward pass of the input.

1) $\mathbf{z}^1$, the net weighted inputs into all neurons in the first layer (1)
2) $\mathbf{a}^1$, the activations of neurons in the first layer (1)
3) $\mathbf{z}^2$, the net weighted inputs into all neurons in the second layer (1)
4) $\mathbf{a}^2$, the activations of neurons in the second layer (1)

(ii) After the forward pass has been performed your model parameters need to be updated. Calculate the following

1) $\boldsymbol{\delta}^L$, the errors in the last layer (3)
2) $\boldsymbol{\delta}^1$, the errors in the first hidden layer (2)
3) $\frac{\partial J}{\partial W^1}$, the gradients for weights in the first layer. (1)
4) $\frac{\partial J}{\partial \mathbf{b}^1}$, the gradients for the biases in the first layer. (1)

(iii) Using gradient descent calculate the new parameters for W^1 given that $\eta = 0.1$ (1)

Question 8: Reinforcement Learning [13 marks]

You have a mobile robot in the discrete grid world shown in Figure 4. There are five states (locations on the grid) and two possible actions it can take at each time step by moving West (W) or East (E) provided there is space to move in to. s_1 and s_5 are terminal states with reward values of $+1$ and $+2$ respectively. The non-terminal states have reward values as shown in Figure 4. The robot begins each trial in s_3.

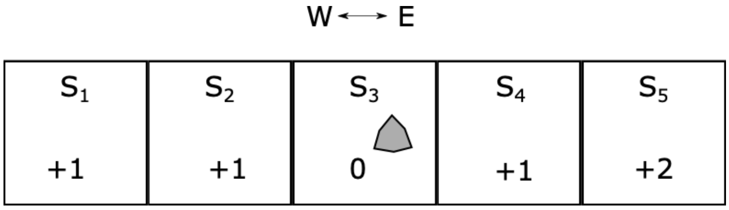


Figure 4: A mobile robot agent in a discrete grid world environment

You decide to use ϵ-greedy Q-learning to train a policy for the agent. All of the initial Q-values are set to zero. The Bellman equation below indicates how the Q-values are updated.

$$Q(s,a) \longleftarrow Q(s,a) + \alpha[R(s) + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

The learning rate is a fixed value given by $\alpha = 0.5$. The discount factor is set to $\gamma = 0.8$. Figure 5 shows the N(s,a) (indicates number of times (s,a) is encountered) and Q(s,a) 'tables' after the first trial. The relative positions of the values in each state cell indicate the action chosen e.g. The number left of the cell corresponds to 'W' and to the right is 'E'. This is to make visualising the transitions easier.

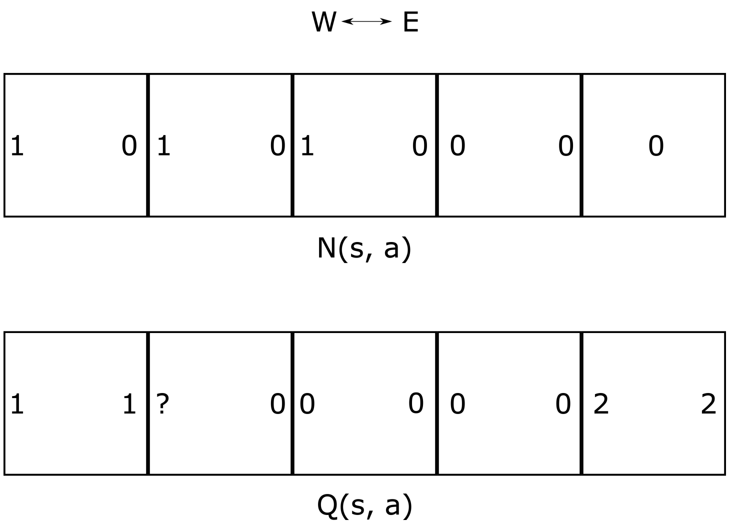


Figure 5: The N(s,a) and Q(s,a) tables after the first trial

a. Calculate the value for $Q(s_2, W)$ after the first trial as indicated in Figure 5 (2)

b. Suppose after the second trial the N(s,a) table is given by Figure 6
   (i) Write down the series of (state, reward, actions) taken that produced the N(s,a) table. (1)
   (ii) Draw the corresponding Q(s,a) table (4)

c. After the second trial, would the agent prefer to move W, E or have no preference? (2)

d. Suppose the third trial takes place as follows: $(s_3, 0, E) \longrightarrow (s_4, 1, E) \longrightarrow (s_5, 2)$.
   (i) Calculate the Q(s,a) value for $Q(s_3, E)$ (2)

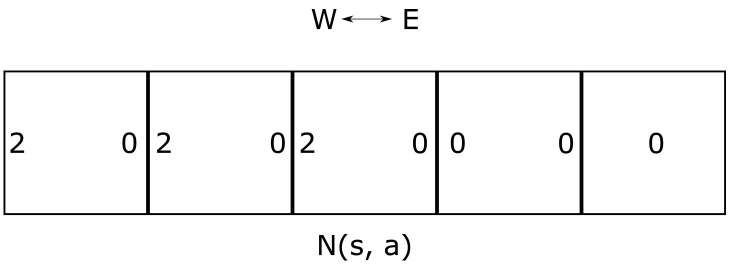


Figure 6: The N(s,a) table for t_2

(ii) After the above trials, would the agent prefer to move W, E or have no preference? Explain your answer. (2)

END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt \iff x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t}d\Omega \quad \text{(CTFT)}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \iff x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} \quad \text{(DTFT)}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \iff x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz \quad \text{(ZT)}$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \iff x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} \quad \text{(DFT)}$$

Discrete-time Fourier transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(e^{j\omega})$, $Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d}X(e^{j\omega})$	Time shift
$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$1 \quad (-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{1-ae^{-j\omega}}$
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{(1-ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \lvert\omega\rvert < \omega_c \\ 0 & \omega_c < \lvert\omega\rvert \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)}e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(z)$, $Y(z)$	ROC	Property
$ax[n]+by[n]$	$aX(z)+bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n-n_d]$	$z^{-n_d}X(z)$	ROC $= R_x$	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC $= \|z_0\|R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC $= \frac{1}{R_x}$	Time reversal
$nx[n]$	$-z\frac{dX(z)}{dz}$	ROC $= R_x$	Frequency diff.
$x[n]*y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC $= R_x$	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\|z\| < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
$\begin{cases} a^n & 0 \le n \le N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\|z\| > 0$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
$r^n\cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\|z\| > \|r\|$

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for} \quad a \neq 1.$$